Supplementary Methods and Results

Forecastability Limits of a P300-Derived EEG Proxy in BCIAUT-P300

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These companion materials are separate from the six-page conference manuscript.

These materials expose all 17 evaluated variants, the original tested contrasts, subject-level differences, ranking aggregation, and existing robustness runs. The original 5–25-block benchmark uses saved predictions and subject-fold metrics without retraining. An additional five-model analysis at horizons 1–4 fitted learned models anew and added 32 direct context-mean comparisons. Its results and inference family are reported separately. The supplement generator reads these completed outputs and fits only the new affine calibration correction described in Section X. Bootstrap intervals are descriptive and unadjusted.

I. Exact proxy construction and annotation requirements

Each preprocessed epoch has eight channels and 128 time samples. The target and non-target prototypes P_{T} and P_{NT} average their corresponding epochs in avatar-cued calibration blocks 10–19. Let $\bar{E}_b$ average the task-designated target epochs in block b. The augmented representation is

$$X_b = [P_{\mathrm{T}}; P_{\mathrm{NT}}; \bar{E}_b], \qquad \widetilde{C}_b = \mathrm{OAS}(X_b^\top). \tag{S1}$$

This is a 24×128 matrix; semicolons denote vertical concatenation. OAS treats the 128 time points as observations and 24 augmented channels as variables; it estimates and removes each variable's temporal mean (`assume_centered=False`) [1]. This specifies the estimator's input convention; it does not imply independent ERP time samples.

The unaligned score references are

$$R_{\mathrm{T}} = \mathrm{OAS}([P_{\mathrm{T}}; P_{\mathrm{NT}}; P_{\mathrm{T}}]^\top), \qquad R_{\mathrm{NT}} = \mathrm{OAS}([P_{\mathrm{T}}; P_{\mathrm{NT}}; P_{\mathrm{NT}}]^\top). \tag{S2}$$

They substitute each calibration prototype for the block response. These deterministic endpoints occupy the same augmented space as X_b and define a prototype-proximity margin without estimating references from future online blocks. They are not empirical Riemannian class means. The augmented ERP representation is inspired by Barachant and Congedo [2]; their decoding method does not independently validate this duplicated-prototype, block-averaged proxy.

The affine-invariant distance and session calibration mean are

$$d_R(A, B) = \| \log(A^{-1/2} B A^{-1/2}) \|_F, \qquad G = \arg\min_{G \succ 0} \sum_{j=0}^{19} d_R^2(G, \widetilde{C}_j). \tag{S3}$$

All 20 calibration blocks enter G [3]. For $W = G^{-1/2}$, the aligned block covariance is $C_b = W \widetilde{C}_b W^\top$ and aligned references are $C_k^{\mathrm{ref}} = W R_k W^\top$, for $k \in \{\mathrm{T}, \mathrm{NT}\}$. The score is $s_b = d_R(C_b, C_{\mathrm{NT}}^{\mathrm{ref}}) - d_R(C_b, C_{\mathrm{T}}^{\mathrm{ref}})$. The tangent matrix is $\log(C_b)$, vectorized in row-wise upper-triangle order with off-diagonal entries multiplied by $\sqrt{2}$ and diagonal entries unchanged, yielding 300 features.

For the same invertible W, $d_R(WAW^\top, WBW^\top) = d_R(A, B)$ [4]. Whitening prepares tangent features for modelling; it does not itself remove session variation from the scalar margin. Session-specific prototypes, reference geometry, calibration fit, and averaging precision can contribute to the score's scale and stability.

The analysis uses released target annotations retrospectively. An avatar-task controller can know the instructed target and each flashed object's identity. Prospective computation requires synchronized cue/stimulus metadata. The instructed target does not establish actual participant attention. The original adaptive ERP method also discusses supervised online target metadata [2]. Attention or engagement interpretation requires independent behavioural validation.

II. Training, inference, and descriptive diagnostics

A. Training settings and existing robustness runs

The main GRUs have one recurrent layer and Gaussian mean/scale outputs. Hidden dimension is selected from $\{32, 64\}$ and learning rate from $\{10^{-3}, 5 \times 10^{-4}\}$; dropout is 0.2 and weight decay is 10^{-4}. EEG-containing GRUs use 20 PCA components fitted on current training subjects. Batch size is 64, the optimizer is Adam, and the main benchmark uses at most 200 epochs with early-stopping patience 20 and no gradient clipping. The short-horizon analysis uses the same historical point-forecasting implementation, preserved in commit `48de5b0` and included as `src/experiments/exp_historical_step.py`. Selection minimizes validation Gaussian NLL, while the primary point metric is MAE; these runs do not exhaust GRUs selected to optimize MAE.

The portable repository's `pipeline/reproduce_paper_gru.py` command restores these historical stopping settings and disables gradient clipping for the main and robustness comparisons. The general Step 8 entrypoint retains later defaults of 100 epochs, patience 10, and gradient clipping; it is not the paper-training command. Validation training and final refitting are compared with the preserved historical implementation in synthetic tests.

The last two outer-training subjects by sorted identifier provide main inner validation. A fresh model is refitted on all 14 training subjects for the best validation epoch. That validation pair is repeated in most outer folds. Archived `exp30` instead uses five-fold inner GroupKFold by subject, selecting by validation-size-weighted mean best NLL. It refits for the rounded mean of the selected candidate's five best epochs, with a minimum of one. Archived `exp31` uses the same grouped protocol and adds dropout $\{0, 0.2\}$, yielding eight hyperparameter combinations.

Tasks are enumerated by horizon, target view, ablation, and held-out subject. For zero-based task index i, candidate g on inner split j uses seed $20260330 + 101i + g\,n_{\text{inner}} + j$; final refitting uses $20260330 + 101i + 97$. PyTorch and the shuffled DataLoader use these seeds. Deterministic algorithms are requested with `warn_only=True`; cross-platform bitwise reproducibility is not asserted. Model-selection CSVs record selected settings and refit epochs.

B. Point metrics, original contrasts, and intervals

For held-out subject u at horizon h, mean absolute error is

$$\text{MAE}_{u,h} = \frac{1}{n_{u,h}} \sum_{i=1}^{n_{u,h}} |y_i - \hat{y}_i|, \tag{S4}$$

where y_i is the observed target, $\hat{y}_i$ is its prediction, and $n_{u,h}$ is the number of valid forecasting examples. Reported MAE averages these 15 subject values with equal weights. For anchor block b, the context mean is $m_b = \frac{1}{10} \sum_{j=0}^{9} s_{b-j}$. Residual fitting uses $r_{b,h} = s_{b+h} - m_b$; adding m_b to a residual forecast reconstructs the raw-score forecast without changing its absolute error.

Seven classical and ten GRU contrasts are listed in Tables S5–S6. Each is evaluated for MAE and Gaussian NLL, both views, and five horizons: $(7 + 10) \times 2 \times 2 \times 5 = 340$ comparisons. Holm correction [5] applies separately within each horizon, view, metric, and classical/GRU family: 40 correction families. These are selected contrasts, not exhaustive direct baseline comparisons. The archived CSV retains zero-handling, p-value method, effects, original bootstrap intervals, and adjusted p-values. Three GRU templates use the best of nine trained classical candidates, selected separately by the lowest aggregate outer-subject mean of the tested metric (ties use the other metric, then model name). That post-hoc comparator selection makes those contrasts exploratory. The 20 horizon/view/metric selections are exposed in a separate CSV; the 17 templates instantiate 23 distinct ordered model pairs. Signed-rank interpretation assumes independent subject differences with approximately symmetric distributions; shared LOSO training makes independence an approximation.

New 95% intervals resample paired subject differences or subject ranking aggregates 10,000 times, with NumPy seed 20261007, using the 2.5th and 97.5th percentiles of the mean. They are descriptive, unadjusted intervals, not new tests or simultaneous intervals. They condition on fitted predictions, omit repeated-training/model-selection uncertainty, and inherit the shared-fold independence limitation. For the main benchmark, formal statements use only the original contrasts; the additional fixed-baseline tests below form a separate post-hoc family.

The main paper's paired-t approximation estimates the mean MAE improvement detectable with approximately 80% power in a two-sided, uncorrected test at $\alpha = 0.05$. It uses the observed standard deviation of the 15 subject-fold differences and assumes independent, approximately normal differences. Shared LOSO training makes independence an approximation. For the raw score-only GRU versus context mean, δ_{80} is approximately 0.0040 at horizons 5 and 10 and 0.0087 at horizon 25 (2.3–4.9% of baseline MAE). These values are rough sensitivity estimates, not power calculations for the Holm-corrected signed-rank analysis or evidence against smaller gains.

C. Ranking aggregation and probabilistic scores

Let τ be 80% of the session's avatar-cued calibration mean. All 105 means are positive: minimum 1.279, first quartile 2.022, median 2.293, third quartile 2.479, maximum 3.280. Thus the signed-margin threshold has no nonpositive or near-zero calibration references in this sample. The underlying score remains signed and unvalidated behaviourally. The avatar-cued blocks also contribute to the prototypes, so this is an in-sample calibration reference, not an independently validated threshold. It uses no future online blocks; a cross-fitted calibration threshold remains an untested sensitivity analysis. The low-proxy endpoint and its Gaussian forecast probability are

$$D_{b,h} = \mathbf{1}\{s_{b+h} < \tau\}, \qquad p(D_{b,h} = 1) = \Phi\left(\frac{\tau - \mu_{\text{raw}}}{\sigma}\right), \tag{S5}$$

where Φ is the standard normal cumulative distribution function and σ is the forecast standard deviation. Raw-score and residual GRUs are fitted separately, with

$$\mu_{\text{raw}} = \begin{cases} \hat{\mu}, & \text{raw-score fit,} \\ m_b + \hat{\mu}_r, & \text{residual fit.} \end{cases} \tag{S6}$$

Each fit learns its own σ. In residual coordinates, the same event is $r_{b,h} < \tau - m_b$, so the endpoint labels do not change. The context-mean comparator has $\mu_{\text{raw}} = m_b$ in both views and the same training-fold residual standard deviation for σ. Matching predictions by subject, session, horizon, and block therefore gives identical comparator probabilities up to numerical precision.

Pooled ROC-AUC combines all eligible examples from a held-out subject's seven sessions, then gives subjects equal weight. Positive–negative pairs from different sessions contribute. Average precision uses the scikit-learn average-precision function, rather than trapezoidal integration. The session-restricted diagnostic computes ROC-AUC and average precision in sessions with both endpoint classes, averages sessions within subjects, then averages subjects. At horizon 5, 103 sessions are eligible; two have no low-proxy examples. All 15 subjects retain eligible sessions. Both predictors and both metrics use the same eligible-session set. No new significance test is supplied. Intervals resample subject aggregates and describe a different estimand from pooled AUC.

Gaussian NLL is $\frac{1}{2}\log(2\pi) + \log\sigma + (y-\mu)^2/(2\sigma^2)$ (lower is better); a density-based NLL can be negative. Nominal 80% Gaussian width is $2\Phi^{-1}(0.9)\sigma$. NLL, width, and coverage are averaged within subjects, then across subjects. Coverage alone establishes neither conditional calibration nor sharpness.

D. Variance decomposition and block horizons

For balanced scores x_{ujb} from 15 subjects, seven sessions, and 50 blocks, let $\bar{x}_u$, $\bar{x}_{uj}$, and $\bar{x}$ be subject, session, and grand means. The descriptive population-variance partition is

$$V_{\text{sub}} = \frac{1}{15}\sum_{u}(\bar{x}_u - \bar{x})^2, \tag{S7}$$

$$V_{\text{sess}} = \frac{1}{105}\sum_{u,j}(\bar{x}_{uj} - \bar{x}_u)^2, \tag{S8}$$

$$V_{\text{within}} = \frac{1}{5250}\sum_{u,j,b}(x_{ujb} - \bar{x}_{uj})^2. \tag{S9}$$

These sum to empirical population variance (denominator N, not $N-1$); they are not fitted random-effects components. The full-session mean includes future observations and is only a retrospective diagnostic. Its context-mean correlation partly reflects shared observations.

Block units align forecasts with completed task blocks. Eight 100 ms flashes with 200 ms gaps give $2.4K$ seconds of stimulation per block, for $K \in [3, 10]$. Ignoring cue, feedback, and pauses, five blocks span 36–120 seconds of stimulation. Five evenly spaced offsets, 5 through 25 blocks, were selected to compare increasingly distant forecasts. This is a benchmark grid, not an intervention-latency or clinical prescription. The additional five-model analysis evaluates horizons 1–4 separately.

III. Complete model results

Tables S1 and S2 show all 17 variants, with equal-subject mean MAE. The lowest residual mean can differ from the context mean without establishing a reliable improvement in a tested contrast. Residual *EEG-only* denotes the learned correction's inputs; reconstructing a raw forecast still adds the score context mean. The companion CSV includes subject SDs, RMSE, paired differences, and original direct-baseline-test status.

TABLE S1
All 17 raw-score variants. Equal-subject mean MAE; bold denotes the lowest descriptive mean, not a significance claim.

Model	5	10	15	20	25
Context mean	**0.1746**	**0.1741**	**0.1767**	**0.1773**	**0.1752**
Persistence	0.2320	0.2256	0.2318	0.2400	0.2414
Moving average (5)	0.1829	0.1824	0.1829	0.1868	0.1869
Training mean	0.3431	0.3435	0.3427	0.3403	0.3383
AR ridge (5)	0.1828	0.1816	0.1827	0.1860	0.1852
GRU score	0.1780	0.1772	0.1814	0.1815	0.1809
EEG ridge	0.3656	0.3633	0.3607	0.3552	0.3527
EEG elastic-net	0.3406	0.3400	0.3378	0.3352	0.3325
EEG PCR	0.3423	0.3439	0.3432	0.3394	0.3365
EEG PLS	0.3579	0.3582	0.3587	0.3538	0.3532
Riemannian KNN	0.3783	0.3749	0.3676	0.3743	0.3611
GRU EEG	0.3365	0.3378	0.3332	0.3258	0.3404
Permuted EEG PCR	0.3423	0.3439	0.3428	0.3396	0.3365
Hybrid ridge	0.1895	0.1878	0.1933	0.1953	0.1943
Hybrid PCR	0.2970	0.2954	0.2972	0.2990	0.3020
Hybrid PLS	0.2561	0.2601	0.2672	0.2622	0.2695
GRU hybrid	0.1922	0.1938	0.1924	0.1930	0.1898

TABLE S2
All 17 separately fitted residual-target variants. Equal-subject mean MAE; bold denotes the lowest descriptive mean, not a significance claim.

Model	5	10	15	20	25
Context mean	0.1746	0.1741	**0.1767**	0.1773	0.1752
Persistence	0.2320	0.2256	0.2318	0.2400	0.2414
Moving average (5)	0.1829	0.1824	0.1829	0.1868	0.1869
Training mean	0.1748	0.1745	0.1772	0.1776	0.1752
AR ridge (5)	0.1747	0.1740	0.1771	**0.1769**	0.1747
GRU score	0.1752	0.1743	0.1775	0.1777	0.1753
EEG ridge	0.1792	0.1787	0.1822	0.1844	0.1770
EEG elastic-net	0.1747	0.1745	0.1772	0.1772	**0.1747**
EEG PCR	0.1747	0.1737	0.1785	0.1787	0.1759
EEG PLS	0.1800	0.1802	0.1823	0.1829	0.1856
Riemannian KNN	0.1911	0.1937	0.1929	0.1944	0.1924
GRU EEG	0.1744	0.1744	0.1789	0.1798	0.1775
Permuted EEG PCR	0.1751	0.1746	0.1775	0.1790	0.1759
Hybrid ridge	0.1780	0.1777	0.1823	0.1818	0.1765
Hybrid PCR	0.1743	**0.1735**	0.1782	0.1791	0.1760
Hybrid PLS	0.1796	0.1797	0.1820	0.1820	0.1847
GRU hybrid	**0.1737**	0.1744	0.1782	0.1782	0.1762

IV. Paired differences from the primary comparator

Positive differences mean worse MAE than the context mean; negative values mean lower descriptive MAE. Tables S3 and S4 show horizons 5 and 25. All five horizons and individual subject differences are in the CSVs. A dash in the final column means that direct comparison was outside the original contrast family. No significance decision is inferred from the unadjusted intervals. The companion paired-difference plot in the release package uses the same interval procedure. Main-paper Figure 5 contrasts retrospective lag correlations with raw- and residual-target GRU MAE differences and these descriptive intervals.

TABLE S3

Raw-target MAE differences (model minus context mean), with unadjusted descriptive 95% subject-fold bootstrap intervals. The final column identifies an original direct baseline test.

Model	$h = 5$: difference [CI]	$h = 25$: difference [CI]	Direct test
Persistence	+0.0575 [+0.0492, +0.0651]	+0.0662 [+0.0595, +0.0725]	Yes
Moving average (5)	+0.0084 [+0.0062, +0.0104]	+0.0117 [+0.0086, +0.0147]	–
Training mean	+0.1686 [+0.1199, +0.2221]	+0.1631 [+0.1165, +0.2134]	–
AR ridge (5)	+0.0082 [+0.0057, +0.0109]	+0.0099 [+0.0065, +0.0135]	–
GRU score	+0.0034 [+0.0011, +0.0062]	+0.0057 [+0.0003, +0.0113]	Yes
EEG ridge	+0.1911 [+0.1483, +0.2400]	+0.1775 [+0.1372, +0.2243]	–
EEG elastic-net	+0.1660 [+0.1199, +0.2171]	+0.1573 [+0.1151, +0.2032]	–
EEG PCR	+0.1677 [+0.1203, +0.2208]	+0.1613 [+0.1183, +0.2069]	Yes
EEG PLS	+0.1833 [+0.1375, +0.2370]	+0.1779 [+0.1338, +0.2279]	–
Riemannian KNN	+0.2037 [+0.1731, +0.2335]	+0.1858 [+0.1528, +0.2175]	–
GRU EEG	+0.1620 [+0.1251, +0.2077]	+0.1651 [+0.1272, +0.2113]	Yes
Permuted EEG PCR	+0.1678 [+0.1205, +0.2208]	+0.1613 [+0.1184, +0.2068]	–
Hybrid ridge	+0.0150 [+0.0095, +0.0216]	+0.0191 [+0.0148, +0.0233]	Yes
Hybrid PCR	+0.1224 [+0.0931, +0.1602]	+0.1268 [+0.0970, +0.1618]	Yes
Hybrid PLS	+0.0815 [+0.0638, +0.1054]	+0.0942 [+0.0765, +0.1139]	–
GRU hybrid	+0.0177 [+0.0071, +0.0315]	+0.0145 [+0.0058, +0.0242]	Yes

TABLE S4

Residual-target MAE differences (model minus context mean), with unadjusted descriptive 95% subject-fold bootstrap intervals. The final column identifies an original direct baseline test.

Model	$h = 5$: difference [CI]	$h = 25$: difference [CI]	Direct test
Persistence	+0.0575 [+0.0492, +0.0651]	+0.0662 [+0.0595, +0.0725]	Yes
Moving average (5)	+0.0084 [+0.0062, +0.0104]	+0.0117 [+0.0086, +0.0147]	–
Training mean	+0.0003 [+0.0001, +0.0005]	-0.0000 [-0.0012, +0.0012]	–
AR ridge (5)	+0.0002 [-0.0002, +0.0005]	-0.0005 [-0.0018, +0.0009]	–
GRU score	+0.0006 [-0.0002, +0.0015]	+0.0001 [-0.0012, +0.0014]	Yes
EEG ridge	+0.0047 [+0.0030, +0.0068]	+0.0018 [-0.0009, +0.0047]	–
EEG elastic-net	+0.0001 [-0.0002, +0.0005]	-0.0005 [-0.0025, +0.0014]	–
EEG PCR	+0.0002 [-0.0012, +0.0019]	+0.0007 [-0.0007, +0.0023]	Yes
EEG PLS	+0.0055 [+0.0016, +0.0098]	+0.0104 [+0.0056, +0.0153]	–
Riemannian KNN	+0.0165 [+0.0130, +0.0203]	+0.0171 [+0.0126, +0.0219]	–
GRU EEG	-0.0001 [-0.0015, +0.0012]	+0.0023 [-0.0011, +0.0062]	Yes
Permuted EEG PCR	+0.0005 [+0.0002, +0.0009]	+0.0007 [-0.0008, +0.0024]	–
Hybrid ridge	+0.0035 [+0.0020, +0.0049]	+0.0013 [-0.0013, +0.0042]	Yes
Hybrid PCR	-0.0002 [-0.0017, +0.0015]	+0.0008 [-0.0007, +0.0023]	Yes
Hybrid PLS	+0.0051 [+0.0012, +0.0096]	+0.0095 [+0.0049, +0.0143]	–
GRU hybrid	-0.0009 [-0.0023, +0.0005]	+0.0009 [-0.0012, +0.0033]	Yes

V. Exact original contrast families

Each family is repeated for every horizon, target view, and metric. Archived differences use model B minus model A; the preceding paired tables consistently use alternative minus context mean. All 340 original tests are preserved in `original_tested_contrasts.csv`. In the GRU table, the selected trained comparator is AR ridge (5) for all raw-view comparisons and all Gaussian-NLL comparisons. For residual MAE, it is hybrid PCR at horizons 5 and 10, AR ridge (5) at 15 and 20, and EEG elastic-net at 25. The selection CSV gives every metric-specific instance.

TABLE S5
Original seven classical contrast templates. Archived differences use model B minus model A.

ID	Model A	Model B
C1	Persistence	Moving average (5)
C2	Context mean	Persistence
C3	Context mean	Hybrid ridge
C4	Context mean	EEG PCR
C5	Context mean	Hybrid PCR
C6	EEG PCR	Riemannian KNN
C7	EEG PCR	Permuted EEG PCR

TABLE S6
Original ten GRU contrast templates. Archived differences use model B minus model A.

ID	Model A	Model B
G1	GRU score	Persistence
G2	GRU score	Context mean
G3	GRU score	Selected trained classical
G4	GRU EEG	Persistence
G5	GRU EEG	Context mean
G6	GRU EEG	Selected trained classical
G7	GRU hybrid	Persistence
G8	GRU hybrid	Context mean
G9	GRU hybrid	Selected trained classical
G10	GRU hybrid	GRU score

VI. Ranking, robustness, and uncertainty

Only session-corrected `review_revision/step_anomaly_predictions.csv` is used for ranking; historical main and robustness ranking files are excluded. The two AUC aggregations use different pairs and session weighting. Moderate pooled AUC does not establish within-session temporal discrimination.

TABLE S7
Horizon-5 ROC-AUC, with unadjusted descriptive 95% subject-bootstrap intervals. Session-restricted results average 103 eligible sessions within subjects, then give the 15 subjects equal weight.

Predictor	Fit	Sessions pooled	Within-session
Context mean	Raw	0.6353 [0.5914, 0.6821]	0.4742 [0.4463, 0.5024]
GRU score	Raw	0.6309 [0.5875, 0.6772]	0.4692 [0.4413, 0.4967]
GRU EEG	Raw	0.5010 [0.4730, 0.5307]	0.5004 [0.4750, 0.5288]
GRU hybrid	Raw	0.5984 [0.5695, 0.6284]	0.4806 [0.4651, 0.4964]
Context mean	Residual	0.6353 [0.5914, 0.6821]	0.4742 [0.4463, 0.5024]
GRU score	Residual	0.6324 [0.5893, 0.6789]	0.4712 [0.4430, 0.4991]
GRU EEG	Residual	0.6316 [0.5903, 0.6772]	0.4781 [0.4465, 0.5069]
GRU hybrid	Residual	0.6323 [0.5906, 0.6784]	0.4798 [0.4508, 0.5077]

The robustness table uses existing `exp30` and `exp31` point-forecasting artifacts. Both raw score-only GRUs remain above the context mean at every horizon. All GRUs, views, and horizons are included in `robustness_summary.csv`.

TABLE S8
Saved robustness runs: raw-target score-only GRU mean MAE.

Protocol	5	10	15	20	25
Context mean	0.1746	0.1741	0.1767	0.1773	0.1752
Main GRU	0.1780	0.1772	0.1814	0.1815	0.1809
Grouped inner CV	0.1803	0.1783	0.1830	0.1798	0.1834
Grouped CV + dropout search	0.1753	0.1767	0.1783	0.1782	0.1785

TABLE S9
Horizon-5 Gaussian NLL (lower is better), nominal 80% interval coverage, and mean interval width in proxy-score units.

Predictor	Fit	NLL	Coverage (%)	Width
Context mean	Raw	-0.0529	82.5	0.5852
GRU score	Raw	-0.0702	84.0	0.6168
GRU EEG	Raw	0.6329	72.1	0.9303
GRU hybrid	Raw	0.0049	80.3	0.6184
Context mean	Residual	-0.0529	82.5	0.5852
GRU score	Residual	-0.0909	82.3	0.5828
GRU EEG	Residual	-0.0466	81.6	0.5883
GRU hybrid	Residual	-0.0721	81.7	0.5780

The main raw score-only GRU's horizon-5 NLL advantage does not establish a reliable gain (original Holm-adjusted $p = 0.4586$). All variants' NLL, coverage, and computed Gaussian width are supplied in `probabilistic_summary.csv`.

VII. Additional short-horizon analysis

The five core models are the context mean, persistence, five-score AR ridge, EEG-context PCR, and score-only GRU. The same frozen session-calibrated proxy, ten-block history, separately fitted raw/residual targets, and 15-subject LOSO evaluation are retained. Valid examples per subject are 280, 273, 266, and 259 at horizons 1–4, respectively, across seven sessions. Each horizon has its own learned coefficients, selected settings, and GRU weights. Classical selection uses the original five-fold inner GroupKFold by subject and MAE grids. GRU selection uses the original two-subject validation, Gaussian NLL, four hidden-size/learning-rate candidates, 200-epoch cap, patience 20, and best-epoch refit. The original three-ablation task enumeration determines seed offsets, although only the score-only GRU is evaluated here. The historical implementation reproduces the saved horizon-5 SBJ01 raw GRU selection (64 units, learning rate 0.001, epoch 82), with maximum prediction difference 4.77×10^{-7}.

PCR computes the same standardized, centred principal subspace using an exact symmetric eigen-decomposition, with full-SVD fallback for ill-conditioned inputs. Decompositions are shared across component candidates and target views. Agreement with the original full-SVD estimator was verified in tests, saved horizon-5 outputs, and three completed short-horizon subject fits, at absolute prediction tolerance 10^{-8}.

The four alternatives are compared directly with the context mean at each horizon and target view, yielding 32 paired MAE tests over 15 subject values. Two-sided Wilcoxon tests retain Pratt zero handling. Primary Holm correction covers all 32 new comparisons; four-test correction within each horizon/view is also reported. These are additional families, distinct from the original seven-classical and ten-GRU families. Descriptive paired 95% intervals use 10,000 subject bootstrap resamples with seed 20261008, condition on fitted predictions, and omit refitting uncertainty. Shared LOSO training limits independence. This analysis reuses the original participants and does not constitute an independent validation cohort. No new ranking analysis is added.

Tables S10–S11 show all five models. The context mean has the lowest raw-score mean MAE at each short horizon. Residual EEG PCR has lower descriptive MAE at horizons 2–4; the largest reduction is 0.000748 (0.43%) at horizon 4, with primary Holm-adjusted $p = 0.886$. No alternative establishes a gain in either view.

TABLE S10
Raw-score short-horizon MAE for the five-model extension. Values average 15 held-out subjects equally.

Model	1	2	3	4
Context mean	0.1721	0.1724	0.1736	0.1741
Persistence	0.2284	0.2298	0.2309	0.2306
AR ridge (5)	0.1793	0.1813	0.1830	0.1828
EEG PCR	0.3420	0.3433	0.3429	0.3425
GRU score	0.1736	0.1755	0.1754	0.1748

TABLE S11
Separately fitted residual-target short-horizon MAE for the five-model extension. Values average 15 held-out subjects equally.

Model	1	2	3	4
Context mean	0.1721	0.1724	0.1736	0.1741
Persistence	0.2284	0.2298	0.2309	0.2306
AR ridge (5)	0.1723	0.1726	0.1737	0.1741
EEG PCR	0.1721	0.1723	0.1730	0.1733
GRU score	0.1730	0.1730	0.1740	0.1742

All 32 paired effects are reported below. Negative differences indicate lower MAE than the context mean. Intervals are unadjusted and descriptive; they do not replace the multiplicity-corrected tests or establish equivalence.

TABLE S12

RAW-TARGET SHORT-HORIZON MAE DIFFERENCES (ALTERNATIVE MINUS CONTEXT MEAN). INTERVALS ARE UNADJUSTED DESCRIPTIVE 95% PAIRED-SUBJECT BOOTSTRAPS. PRIMARY p_{32} CORRECTS ALL 32 ADDITIONAL COMPARISONS; p_4 CORRECTS WITHIN EACH HORIZON/VIEW.

Alternative	h	Δ MAE	95% interval	p_{32}	p_4
Persistence	1	+0.05628	[+0.04914, +0.06454]	0.0020	0.0002
Persistence	2	+0.05740	[+0.05207, +0.06278]	0.0020	0.0002
Persistence	3	+0.05731	[+0.05069, +0.06372]	0.0020	0.0002
Persistence	4	+0.05655	[+0.05080, +0.06245]	0.0020	0.0002
AR ridge (5)	1	+0.00721	[+0.00483, +0.00945]	0.0052	0.0006
AR ridge (5)	2	+0.00885	[+0.00648, +0.01121]	0.0022	0.0002
AR ridge (5)	3	+0.00941	[+0.00742, +0.01130]	0.0020	0.0002
AR ridge (5)	4	+0.00875	[+0.00659, +0.01091]	0.0020	0.0002
EEG PCR	1	+0.16994	[+0.12326, +0.22220]	0.0020	0.0002
EEG PCR	2	+0.17083	[+0.12375, +0.22337]	0.0020	0.0002
EEG PCR	3	+0.16932	[+0.12238, +0.22184]	0.0020	0.0002
EEG PCR	4	+0.16845	[+0.12167, +0.22059]	0.0020	0.0002
GRU score	1	+0.00151	[-0.00058, +0.00438]	1.0000	0.6387
GRU score	2	+0.00301	[+0.00008, +0.00760]	1.0000	0.1688
GRU score	3	+0.00184	[-0.00040, +0.00545]	1.0000	0.5614
GRU score	4	+0.00072	[-0.00034, +0.00197]	1.0000	0.3591

TABLE S13

RESIDUAL-TARGET SHORT-HORIZON MAE DIFFERENCES (ALTERNATIVE MINUS CONTEXT MEAN). INTERVALS ARE UNADJUSTED DESCRIPTIVE 95% PAIRED-SUBJECT BOOTSTRAPS. PRIMARY p_{32} CORRECTS ALL 32 ADDITIONAL COMPARISONS; p_4 CORRECTS WITHIN EACH HORIZON/VIEW.

Alternative	h	Δ MAE	95% interval	p_{32}	p_4
Persistence	1	+0.05628	[+0.04914, +0.06454]	0.0020	0.0002
Persistence	2	+0.05740	[+0.05207, +0.06278]	0.0020	0.0002
Persistence	3	+0.05731	[+0.05069, +0.06372]	0.0020	0.0002
Persistence	4	+0.05655	[+0.05080, +0.06245]	0.0020	0.0002
AR ridge (5)	1	+0.00025	[-0.00020, +0.00090]	1.0000	1.0000
AR ridge (5)	2	+0.00015	[-0.00033, +0.00080]	1.0000	1.0000
AR ridge (5)	3	+0.00013	[-0.00037, +0.00079]	1.0000	0.8040
AR ridge (5)	4	+0.00003	[-0.00040, +0.00054]	1.0000	1.0000
EEG PCR	1	+0.00000	[-0.00091, +0.00105]	1.0000	1.0000
EEG PCR	2	-0.00017	[-0.00111, +0.00085]	1.0000	1.0000
EEG PCR	3	-0.00055	[-0.00148, +0.00057]	1.0000	0.4061
EEG PCR	4	-0.00075	[-0.00191, +0.00079]	0.8857	0.1661
GRU score	1	+0.00088	[-0.00028, +0.00232]	1.0000	1.0000
GRU score	2	+0.00060	[-0.00031, +0.00181]	1.0000	1.0000
GRU score	3	+0.00043	[-0.00013, +0.00109]	1.0000	0.5537
GRU score	4	+0.00016	[-0.00046, +0.00085]	1.0000	1.0000

VIII. Additional fixed baselines

Two post-hoc fixed predictors test whether a different estimate of the session level improves forecasting. The calibration mean predicts every future score with the session's mean from avatar-cued blocks 10–19. The expanding online mean predicts with $\bar{s}_{0:b} = (b+1)^{-1} \sum_{j=0}^{b} s_j$. Both use only information available at the observed anchor and retain the same valid anchors, targets, and held-out subjects as the original experiments. The expanding mean uses earlier observed blocks outside the original ten-block context, so this is a comparison with a longer-history fixed rule, not an identical-input model comparison. Neither rule needs training or a selected hyperparameter.

The nine horizons supply 29,820 distinct forecasting examples. MAE is averaged within each subject, then equally across all 15 subjects. The fixed forecasts are evaluated once on the raw-score scale: subtracting the context mean from both target and forecast would duplicate the same comparison. The original context-mean subject MAEs are reproduced at absolute tolerance 10^{-12}.

TABLE S14
Additional fixed baselines on the original valid examples. MAE gives 15 subjects equal weight; the expanding mean uses all online blocks observed through the anchor.

Horizon	Context mean	Calibration mean	Expanding mean
1	0.1721	0.4517	0.1691
2	0.1724	0.4517	0.1690
3	0.1736	0.4518	0.1697
4	0.1741	0.4519	0.1700
5	0.1746	0.4523	0.1701
10	0.1741	0.4535	0.1710
15	0.1767	0.4552	0.1720
20	0.1773	0.4623	0.1723
25	0.1752	0.4611	0.1739

The expanding mean has lower average MAE at every horizon. At horizon 5, MAE decreases by 0.00443 (2.54%); at 25, it decreases by 0.00128 (0.73%). The calibration mean performs substantially worse despite a high association between session means. Online means average 0.444 below the avatar-calibration means. Correlation does not remove this level bias. The all-calibration/online mean correlation is 0.958; using avatar-cued calibration only gives 0.946. The session-level CSV distinguishes these two definitions explicitly.

Each additional baseline is compared with the context mean at each horizon, giving 18 new two-sided paired Wilcoxon tests over 15 subject MAEs, with Pratt zero handling and exact signed-rank inference. A separate Holm family covers all 18 comparisons. This post-hoc analysis does not modify the original 340 tests or the short-horizon extension's 32-test family. Differences favouring the expanding mean survive correction at horizons 1–5, 10, 15, and 20, but not 25 ($p_{18} = 0.303$).

The descriptive paired intervals resample 15 subjects 10,000 times with NumPy seed 20261011. They condition on the observed scores; the shared LOSO training limitation applies to comparisons involving learned fits, and overlapping forecast windows do not supply additional independent subjects. These additional results reuse the original participants and do not constitute independent confirmation. A calibration-context blend and MAE-trained GRU are not evaluated.

TABLE S15
ADDITIONAL POST-HOC PAIRED MAE CONTRASTS (ALTERNATIVE MINUS CONTEXT MEAN). INTERVALS ARE UNADJUSTED DESCRIPTIVE 95% SUBJECT BOOTSTRAPS. p_{18} CORRECTS ALL 18 NEW TWO-SIDED TESTS.

Alternative	h	Δ MAE	95% interval	p_{18}
Calibration mean	1	+0.27960	[+0.25299, +0.30600]	0.0011
Expanding mean	1	-0.00298	[-0.00394, -0.00194]	0.0027
Calibration mean	2	+0.27926	[+0.25282, +0.30538]	0.0011
Expanding mean	2	-0.00343	[-0.00481, -0.00206]	0.0037
Calibration mean	3	+0.27820	[+0.25112, +0.30475]	0.0011
Expanding mean	3	-0.00387	[-0.00558, -0.00211]	0.0058
Calibration mean	4	+0.27779	[+0.25098, +0.30424]	0.0011
Expanding mean	4	-0.00407	[-0.00598, -0.00218]	0.0079
Calibration mean	5	+0.27777	[+0.25094, +0.30442]	0.0011
Expanding mean	5	-0.00443	[-0.00635, -0.00248]	0.0058
Calibration mean	10	+0.27945	[+0.25258, +0.30575]	0.0011
Expanding mean	10	-0.00310	[-0.00454, -0.00168]	0.0079
Calibration mean	15	+0.27852	[+0.25252, +0.30441]	0.0011
Expanding mean	15	-0.00469	[-0.00608, -0.00324]	0.0027
Calibration mean	20	+0.28508	[+0.25764, +0.31304]	0.0011
Expanding mean	20	-0.00497	[-0.00701, -0.00298]	0.0030
Calibration mean	25	+0.28591	[+0.26188, +0.31033]	0.0011
Expanding mean	25	-0.00128	[-0.00359, +0.00106]	0.3028

IX. Continuous residual-prediction diagnostic

For each of the 13 learned variants, both fitted target views, and five main horizons, the diagnostic compares observed context residuals $r_{b,h} = s_{b+h} - m_b$ with predicted residuals. For residual-target fits, $\hat{r}_{b,h}$ is the stored model output; for raw-target fits it is $\hat{s}_{b+h} - m_b$. Saved predictions are matched to scores by subject, session, horizon, anchor, and target. Archived targets and all 1,950 learned-model subject MAEs are independently reproduced to 10^{-12}. For association calculations, the recent mean and observed residual are computed in float64 from the saved float32 scores; archived targets retain their original float32 rounding for the MAE verification.

Pearson correlation is computed separately within each session's valid anchors, then correlations are averaged within subjects and subjects equally. This avoids a pooled association driven by between-session level differences. Sessions with predicted or observed SD at most 10^{-12} have undefined correlation and are excluded; the CSV reports eligibility. Residual elastic-net has constant predictions at horizons 10 and 15 (shown as a dash), and only 13 subjects/91 sessions contribute at 20. All other residual-target entries use 15 subjects and 105 sessions. The zero context-mean correction is also constant, so no baseline correlation is defined.

TABLE S16

Within-session Pearson associations between predicted and observed context residuals for separately fitted residual-target models. Session correlations are averaged within subjects, then subjects equally. Subject uncertainty is in the companion CSV.

Residual-target model	5	10	15	20	25
AR ridge (5)	+0.172	+0.114	+0.124	+0.146	+0.081
EEG ridge	+0.013	+0.010	-0.004	+0.012	+0.069
EEG elastic-net	+0.060	—	—	+0.051	+0.058
EEG PCR	+0.146	+0.108	+0.026	+0.036	-0.026
EEG PLS	+0.159	+0.113	+0.094	+0.109	+0.036
Riemannian KNN	+0.011	-0.006	+0.024	+0.026	+0.018
Permuted EEG PCR	-0.005	+0.015	-0.013	-0.034	-0.011
Hybrid ridge	+0.051	+0.042	+0.014	+0.036	+0.071
Hybrid PCR	+0.160	+0.115	+0.072	+0.040	-0.027
Hybrid PLS	+0.166	+0.118	+0.100	+0.115	+0.040
GRU score	+0.182	+0.112	+0.095	+0.136	+0.091
GRU EEG	+0.173	+0.161	+0.138	+0.112	+0.049
GRU hybrid	+0.197	+0.179	+0.162	+0.136	+0.069

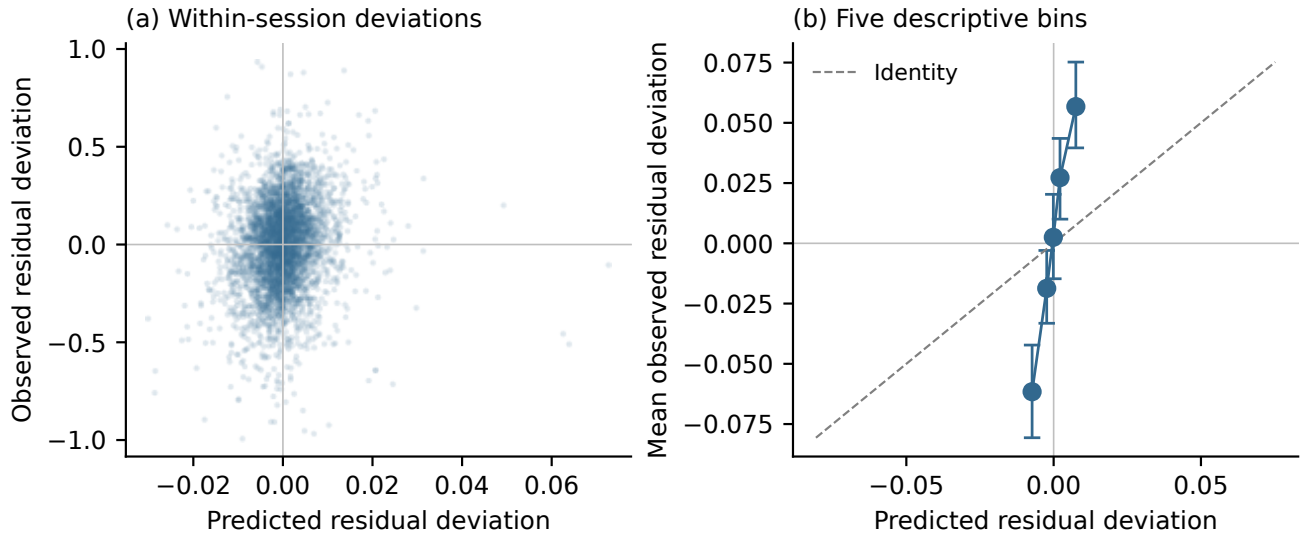


Fig. S1. Residual-target score-only GRU at horizon 5. Session means are subtracted from predicted and observed residuals for this retrospective display, not supplied to the forecaster. (a) All 3,780 held-out pairs. (b) Five pooled predicted-deviation quantile bins; bin means give subjects equal weight and bars show unadjusted descriptive 95% subject-bootstrap intervals. Bins and centring are display diagnostics, not training decisions. The identity line indicates matching magnitudes.

TABLE S17

RESIDUAL-TARGET GRU ASSOCIATIONS WITH UNADJUSTED DESCRIPTIVE 95% SUBJECT-BOOTSTRAP INTERVALS. SD RATIO AVERAGES WITHIN-SESSION PREDICTED/OBSERVED RESIDUAL SDS WITHIN SUBJECTS AND THEN ACROSS SUBJECTS; IT DESCRIBES VARIATION, NOT FORECAST CALIBRATION.

Model	h	Pearson r	95% interval	SD ratio
GRU EEG	5	+0.173	[+0.151, +0.196]	0.121
GRU EEG	10	+0.161	[+0.122, +0.198]	0.141
GRU EEG	15	+0.138	[+0.089, +0.186]	0.181
GRU EEG	20	+0.112	[+0.082, +0.143]	0.175
GRU EEG	25	+0.049	[-0.003, +0.098]	0.166
GRU hybrid	5	+0.197	[+0.176, +0.217]	0.126
GRU hybrid	10	+0.179	[+0.137, +0.222]	0.150
GRU hybrid	15	+0.162	[+0.119, +0.207]	0.169
GRU hybrid	20	+0.136	[+0.097, +0.172]	0.164
GRU hybrid	25	+0.069	[+0.002, +0.133]	0.160
GRU score	5	+0.182	[+0.148, +0.215]	0.023
GRU score	10	+0.112	[+0.067, +0.156]	0.042
GRU score	15	+0.095	[+0.048, +0.143]	0.026
GRU score	20	+0.136	[+0.082, +0.190]	0.029
GRU score	25	+0.091	[+0.045, +0.133]	0.045

Intervals use 10,000 subject bootstrap resamples and seed 20261011, condition on saved fits, and omit refitting/model-selection uncertainty. They are unadjusted, descriptive intervals; no residual-correlation significance tests or equivalence claims are added. Shared LOSO training, short session series, and overlapping windows limit their interpretation. Full-session centring can affect finite-series lag correlations; the original centred-lag diagnostic does not prove an absence of temporal structure.

Small positive residual associations coexist with limited MAE gains. At horizon 5, the residual score-only GRU's mean within-session correlation is 0.182 (95% interval $[0.148, 0.215]$), but its predicted/observed residual SD ratio averages only 0.023. EEG and hybrid residual GRUs give correlations 0.173 and 0.197 and SD ratios 0.121 and 0.126, respectively. The small SD ratio describes compressed predictive variation; it does not alone establish miscalibration, since conditional-mean forecasts can vary less than noisy observations. These associations concern direction and covariation; they do not establish accurate forecast magnitudes, behavioural meaning, or an MAE advantage. Subtracting the recent mean defines the target; it does not isolate a purely physiological dynamic component. Both fitted views, all session metrics, subject aggregates, eligibility counts, and the figure's points/bins are supplied.

X. Training-subject-fitted calibration correction

The strong calibration/online session-mean association and systematic offset motivate one further post-hoc baseline:

$$\hat{s}_{b+h} = a_{-u} + c_{-u}\bar{s}^{\text{avatar}}_{u,j,\text{cal}}. \tag{S10}$$

For fold u, ordinary least squares maps avatar-calibration means to full 50-block online-session means using the other 14 subjects' 98 sessions. Sessions have equal weight (seven per subject). Only training subjects' online outcomes enter fitting. Prediction uses the held-out session's calibration mean, available before online operation; coefficients are shared across horizons and anchors. No blend, clipping, tuning, or alternative correction is selected. The supplied specification, `docs/paper_revision/calibration_correction_protocol.md`, was written before evaluating this correction, after reviewing earlier diagnostics.

TABLE S18
Post-hoc training-subject-fitted affine calibration correction. MAE gives subjects equal weight. Context and expanding means use observed online history; the correction uses only the held-out session's calibration mean.

Horizon	Context mean	Expanding mean	Corrected calibration
1	0.1721	0.1691	0.1946
2	0.1724	0.1690	0.1940
3	0.1736	0.1697	0.1939
4	0.1741	0.1700	0.1938
5	0.1746	0.1701	0.1933
10	0.1741	0.1710	0.1932
15	0.1767	0.1720	0.1925
20	0.1773	0.1723	0.1919
25	0.1752	0.1739	0.1943

Correction substantially improves the uncorrected calibration forecast: at horizon 5, MAE falls from 0.4523 to 0.1933, a 57.3% reduction. Across horizons, corrected MAE is 0.1919–0.1946; both the recent and expanding online means nevertheless have lower MAE at every horizon after correction. The largest adjusted p is 0.0215 versus the context mean and 0.0137 versus the expanding mean. Fold slopes range from 0.8524 to 0.8863 and intercepts from -0.1849 to -0.1075. Calibration therefore contains useful level information after correcting bias, but this mapping does not match estimates based on observed online history. The comparison concerns these predictors, not the value of calibration information in all possible models.

TABLE S19
Post-hoc corrected-calibration contrasts (correction minus comparator). Intervals are unadjusted descriptive 95% subject bootstraps. p_{18} corrects all 18 correction tests in a family separate from Table S15.

Comparator	h	Δ MAE	95% interval	p_{18}
Context mean	1	+0.02252	[+0.01240, +0.03333]	0.0116
Expanding mean	1	+0.02549	[+0.01572, +0.03617]	0.0022
Context mean	2	+0.02158	[+0.01127, +0.03254]	0.0210
Expanding mean	2	+0.02502	[+0.01517, +0.03582]	0.0049
Context mean	3	+0.02029	[+0.01030, +0.03097]	0.0210
Expanding mean	3	+0.02416	[+0.01433, +0.03497]	0.0049
Context mean	4	+0.01972	[+0.00958, +0.03057]	0.0210
Expanding mean	4	+0.02379	[+0.01398, +0.03465]	0.0056
Context mean	5	+0.01872	[+0.00848, +0.02966]	0.0210
Expanding mean	5	+0.02315	[+0.01355, +0.03380]	0.0056
Context mean	10	+0.01911	[+0.00962, +0.02993]	0.0210
Expanding mean	10	+0.02221	[+0.01333, +0.03248]	0.0031
Context mean	15	+0.01574	[+0.00653, +0.02632]	0.0210
Expanding mean	15	+0.02043	[+0.01159, +0.03077]	0.0049
Context mean	20	+0.01463	[+0.00443, +0.02565]	0.0215
Expanding mean	20	+0.01960	[+0.01010, +0.03038]	0.0137
Context mean	25	+0.01905	[+0.00889, +0.03001]	0.0210
Expanding mean	25	+0.02034	[+0.01127, +0.03037]	0.0067

The 18 correction-versus-online-mean contrasts use identical examples and two-sided subject-paired Wilcoxon tests with Pratt zero handling. One all-18 Holm family is separate from Table S15 and historical families. Descriptive intervals use 10,000 subject resamples and seed 20261011, conditional on fits; shared LOSO training and omitted refitting uncertainty remain limitations. The same participants are reused, without independent confirmation. A blend and point-loss GRU are deferred. Saved fold parameters, training-subject lists, 29,820 predictions, 135 subject MAEs, nine summaries, and 18 tests are supplied.

XI. Availability, provenance, and reproduction

The local package is named `forecastability-limits-proxy-bciaut-p300`. The project repository is:

https://github.com/ada-teodorescu/forecastability-limits-proxy-bciaut-p300

The package includes supplementary sources and its PDF, derived results, relevant pipeline code, `pyproject.toml`, `uv.lock`, and a SHA-256 source manifest. The main manuscript is maintained separately. Raw EEG and third-party research-paper PDFs are not redistributed.

`pipeline/build_paper_supplement.py` reads saved inputs, verifies original paired means/differences, reproduces pooled ranking, and regenerates diagnostic results, tables, and the paired-difference figure. It fits only the affine calibration correction and preserves historical result files. An optional external manuscript argument checks all 80 original main-table MAEs and the five expanding-mean values added to Table I. It also verifies the 40 short-horizon MAE entries, their 600 subject metrics, and all 32 new paired comparisons, and regenerates Tables S10–S13. `pipeline/verify_paper_inference.py` separately checks the original 340 tests and Holm corrections. `pipeline/build_paper_feedback_addendum.py` reproduces the two additional fixed baselines, the 18 new paired tests, calibration-mean checks, continuous residual diagnostics, Tables S14–S19, and Figure S1 from saved artifacts, and fits the training-only affine calibration correction with 18 further tests. The supplement builder invokes it automatically; no original classical or GRU model is retrained. Its independent verifier reconstructs all 15 calibration fits via centered covariance and audits the new predictions and inference.

The manifest hashes source/input files, records package versions and bootstrap settings, identifies corrected ranking inputs, and lists excluded historical ranking files. Outputs under `data/forecasting/review_revision/supplementary/` include:

- `full_model_summary.csv`: 170 model/view/horizon summaries.
- `subject_metrics.csv`: 2,550 subject-fold metric rows.
- `paired_mae_differences.csv`: 2,400 paired subject differences.
- `original_tested_contrasts.csv`: 340 original paired tests.
- `contrast_templates.csv`: 17 original contrast templates.
- `contrast_definitions.csv`: 23 instantiated ordered model pairs.
- `selected_classical_comparators.csv`: 20 comparator selections.
- `ranking_session_metrics.csv`: session eligibility and ranking metrics.
- `ranking_subject_metrics.csv`: both subject ranking aggregations.
- `ranking_summary.csv`: ranking summaries and descriptive intervals.
- `robustness_summary.csv`: 90 GRU/view/horizon/protocol summaries.
- `probabilistic_summary.csv`: Gaussian NLL, coverage, and interval width.
- `variance_components.csv`: the constructed score's variance partition.
- `score_diagnostics.json`: score partition and context correlations.
- `snapshot_manifest.json`: source/input hashes and analysis provenance.
- `short_horizon_summary.csv`: 40 additional model/view/horizon summaries.
- `short_horizon_subject_metrics.csv`: 600 short-horizon subject metrics.
- `short_horizon_paired_mae_stats.csv`: 32 new tests, effects, intervals, and both Holm corrections.
- `additional_baseline_predictions.csv`: all 29,820 example keys, observed targets, and the three fixed predictions.
- `additional_baseline_subject_metrics.csv`: 405 subject MAEs.
- `additional_baseline_summary.csv`: 27 baseline/horizon summaries.
- `additional_baseline_paired_stats.csv`: 18 post-hoc paired tests.
- `calibration_correction_fold_parameters.csv`: 15 training-only affine fits.
- `calibration_correction_predictions.csv`: 29,820 held-out forecasts.
- `calibration_correction_subject_metrics.csv`: 135 subject MAEs.
- `calibration_correction_summary.csv`: nine equal-subject summaries.
- `calibration_correction_paired_stats.csv`: 18 further post-hoc tests.
- `calibration_mean_session_diagnostics.csv`: per-session means and bias.
- `calibration_mean_diagnostics.json`: signs, distribution and correlations.
- `residual_association_session_metrics.csv`: within-session associations.
- `residual_association_subject_metrics.csv`: subject aggregates.
- `residual_association_summary.csv`: uncertainty, variation and eligibility.
- `residual_diagnostic_h_points.csv`: Figure S1's held-out pairs.
- `residual_diagnostic_h_bins.csv`: Figure S1's bin means and intervals.
- `additional_analysis_provenance.json`: analysis definitions, hashes and verification settings.
- `additional_analysis_verification.json`: independent audit results.

The completed short-horizon experiment is stored separately under `data/forecasting/review_revision/short_horizons/`, including predictions, selected settings, the locked protocol, source/input hashes, reference checks, and an independent exhaustive signed-rank audit. All 161,700 predictions reproduce their 600 subject MAEs. The inference audit reproduces all 32 raw p-values, both Holm correction schemes, and all bootstrap intervals. `pipeline/exp_paper_short_horizons.py` refits this analysis from the frozen Step 4/5 artifacts. It preserves completed experiment checkpoints; after source or manuscript revisions, use a new output directory.

Large prediction CSVs are supplied as lossless gzip archives. Analysis tools read them directly, preferring an uncompressed CSV if one has been regenerated. The portable release inventory and compressed/uncompressed hashes are in `docs/paper_revision/release_manifest.json`; original thesis manifests are retained separately as historical provenance. The reproduction guide documents rebuilding the analyses without the private manuscript source.

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